

ARVENA v1.0

CONTROL, STATE, ENTITLEMENT & ACCEPTANCE SPECIFICATION

Status: BUILD-READY

1. SYSTEM ROLES

Roles:

OWNER
ADMIN
SALES
WORKER
SUPERVISOR
FINANCE

Ultra may create custom roles.

2. BASE PERMISSION MATRIX

Legend:

- ✓ Full
- ♦ Scoped
- ✗ No access

Capability	Owner	Admin	Sales	Worker	Supervisor	Finance
Dashboard	✓	✓	♦	♦	♦	♦
Customers view	✓	✓	♦	♦	♦	♦
Customers create	✓	✓	✓	✗	♦	✗
Customers edit	✓	✓	♦	✗	♦	✗
Leads view	✓	✓	♦	✗	♦	✗
Leads create	✓	✓	✓	✗	♦	✗
Leads edit	✓	✓	♦	✗	♦	✗

Capability	Owner	Admin	Sales	Worker	Supervisor	Finance
Quotes	✓	✓	♦	✗	♦	♦
Jobs view	✓	✓	♦	♦	♦	♦
Jobs create	✓	✓	♦	✗	♦	✗
Jobs edit	✓	✓	♦	✗	♦	✗
Assign workers	✓	✓	✗	✗	♦	✗
Visits	✓	✓	♦	♦	♦	✗
Worker status update	✓	✓	✗	assigned only	team scope	✗
Assets	✓	✓	♦	assigned only	♦	✗
Invoices view	✓	♦	♦	✗	✗	✓
Invoice create	✓	♦	♦	✗	✗	✓
Invoice issue	✓	♦	✗	✗	✗	✓
Record payment	✓	♦	✗	✗	✗	✓
Reports	✓	♦	♦	✗	♦	✓
Team management	✓	♦	✗	✗	♦	✗
Integrations	✓	✗	✗	✗	✗	✗
Billing	✓	✗	✗	✗	✗	✗
Settings	✓	♦	✗	✗	✗	✗
Audit	✓	✗	✗	✗	✗	✗

3. SCOPED ACCESS DEFINITIONS

Sales:

- assigned leads;
- assigned customers;
- quotes they own or permitted shared pool.

Worker:

- visits assigned to them;
- job/customer information required to execute those visits.

Supervisor:

- configured team/branch scope.

Admin:

- operational organization scope;
- finance and settings limited by explicit permission.

Finance:

- financial customer identity required for billing;
 - no unnecessary operational/private worker controls.
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4. OWNER SAFETY RULE

There must always be at least one active Owner.

System must not allow disabling/removing final active Owner without ownership transfer.

5. LEAD STATE MACHINE

States:

NEW
CONTACTED
QUALIFIED
QUOTE
NEGOTIATION
WON
LOST

Allowed baseline transitions:

NEW → CONTACTED
NEW → LOST

CONTACTED → QUALIFIED
CONTACTED → LOST

QUALIFIED → QUOTE
QUALIFIED → LOST

QUOTE → NEGOTIATION
QUOTE → WON
QUOTE → LOST

NEGOTIATION → WON
NEGOTIATION → LOST

LOST → CONTACTED only through explicit reopen.

WON is terminal operationally but may retain history.

6. QUOTE STATE MACHINE

DRAFT
ISSUED
ACCEPTED
REJECTED
EXPIRED

Transitions:

DRAFT → ISSUED

ISSUED → ACCEPTED
ISSUED → REJECTED
ISSUED → EXPIRED

Accepted: cannot be silently edited.

Change requires new revision/duplicate quote.

7. JOB STATE MACHINE

DRAFT
CONFIRMED
SCHEDULED
IN_PROGRESS
ON_HOLD
COMPLETED
CANCELLED

Transitions:

DRAFT → CONFIRMED

DRAFT → CANCELLED

CONFIRMED → SCHEDULED

CONFIRMED → CANCELLED

SCHEDULED → IN_PROGRESS

SCHEDULED → ON_HOLD

SCHEDULED → CANCELLED

IN_PROGRESS → ON_HOLD

IN_PROGRESS → COMPLETED

IN_PROGRESS → CANCELLED only privileged with reason

ON_HOLD → SCHEDULED

ON_HOLD → IN_PROGRESS

ON_HOLD → CANCELLED

COMPLETED: terminal except privileged reopen.

CANCELLED: terminal except privileged reopen.

8. JOB COMPLETION RULE

A Job is not automatically completed merely because one Visit is completed.

Default completion condition:

- all required Visits completed;
- required checklist satisfied;
- required photos uploaded/safely pending according to business rule;
- explicit Finish Job action or configured auto-complete rule.

9. VISIT STATE MACHINE

SCHEDULED

EN_ROUTE

ARRIVED

IN_PROGRESS

COMPLETED

CANCELLED

NO_SHOW

Transitions:

SCHEDULED → EN_ROUTE
SCHEDULED → CANCELLED
SCHEDULED → NO_SHOW

EN_ROUTE → ARRIVED
EN_ROUTE → CANCELLED

ARRIVED → IN_PROGRESS
ARRIVED → CANCELLED

IN_PROGRESS → COMPLETED

10. INVOICE STATE MACHINE

DRAFT
ISSUED
PARTIALLY_PAID
PAID
OVERDUE
VOID

Transitions:

DRAFT → ISSUED
DRAFT → VOID

ISSUED → PARTIALLY_PAID
ISSUED → PAID
ISSUED → OVERDUE
ISSUED → VOID privileged

PARTIALLY_PAID → PAID
PARTIALLY_PAID → OVERDUE

OVERDUE → PARTIALLY_PAID
OVERDUE → PAID

PAID: cannot return to issued through normal action.

Correction uses payment reversal/adjustment flow later.

11. FOLLOW-UP STATE

PENDING
DONE
CANCELLED

Snooze does not create new logical follow-up unless configured.

It updates due date and records history.

12. AUTOMATION RUN STATE

PENDING
SCHEDULED
RUNNING
SUCCEEDED
FAILED_RETRYABLE
FAILED_FINAL
CANCELLED

Before execution: revalidate current business state.

13. FREE ENTITLEMENTS

ARVENA Free:

Monthly price: Rp0

Limits:

Organizations: 1

Team: Owner + 2 members

Jobs: 30 created/month

Customers: No artificial small count cap

Leads: No small artificial count cap

Storage: 500 MB baseline

Branches: 1

WhatsApp shortcut: Yes

WhatsApp API: No

Automation: No external automatic messaging

Basic recurring: Yes

Quotes: Yes

Invoices: Yes

Payments: Yes

Assets: Basic

Reports: Basic

Import/export: Yes

Custom fields: No

Advanced roles: No

Audit: Basic critical system audit only, not full user-visible audit module.

14. PRO ENTITLEMENTS

Price: Rp300,000/month

Team: up to 10 members baseline

Jobs: 500/month baseline

Storage: 10 GB baseline

Branches: 1

WhatsApp API: Yes

Inbound WA → customer/lead: Yes

Automation: template-based

Recurring: advanced

Custom fields: Yes

Service catalog: Yes

Branded quote/invoice: Yes

Team performance: Yes

Reports: advanced

Admins: multiple

Messaging: usage/credit based

API: No public API baseline

15. ULTRA ENTITLEMENTS

Price: Rp799,000/month

Team: up to 30 members baseline

Jobs: 2,000/month baseline

Storage: 50 GB baseline

Branches: up to 5 baseline

Custom roles: Yes

Advanced automation: Yes

Audit log: Full

Approval flows: Yes

SLA/Ticket: Yes

Advanced assets: Yes

Recurring contracts: Yes

Advanced analytics: Yes

Public API/Webhooks: Yes

Additional usage: add-ons.

16. ENTITLEMENT OVERRIDE

Plan limits stored centrally.

Commercial team may later issue organization-specific overrides.

Frontend and backend read same effective entitlement.

Backend always authoritative.

17. LIMIT UX

Never suddenly block without warning.

At:

80% usage: show gentle warning.

100%: block new cost-generating creation only where required.

Do not block access to existing data.

Example: Job limit reached.

User can still: view/edit/complete existing jobs.

18. NON-FUNCTIONAL REQUIREMENTS — AVAILABILITY

Initial production target: 99.5% service availability excluding scheduled maintenance.

Later target: 99.9%.

Core operation should degrade gracefully when integration provider fails.

19. RTO / RPO

Initial targets:

RPO: ≤ 24 hours for catastrophic platform-level backup recovery.

RTO: ≤ 8 hours initial bootstrap stage.

As paid critical customer base grows:

RPO: ≤ 1 hour where infrastructure supports.

RTO: ≤ 4 hours.

These must be revisited before enterprise expansion.

20. PERFORMANCE

Target:

Dashboard initial meaningful content: < 2.5 seconds typical broadband/mobile connection.

Common API p95: < 800 ms.

Simple mutation: < 1 second server processing when external provider not involved.

Worker status UI: immediate acknowledgement.

21. SCALE TARGET — V1 ARCHITECTURE

Architecture should safely evolve toward:

100k organizations

millions of customers

millions of jobs

without tenant schema redesign.

Do not prematurely provision infrastructure for that load.

22. STORAGE LIMITS

Initial image upload target:

before compression: max 15 MB/image

post-processing preferred: $\leq$ 2–3 MB typical field photo

Accepted baseline: JPEG

PNG

WebP

Document types defined separately.

23. SECURITY REQUIREMENTS

Mandatory:

- HTTPS only;
 - secure cookies/session;
 - RLS;
 - server authorization;
 - tenant constraints;
 - rate limits;
 - webhook signature validation;
 - secrets vault/env;
 - private media;
 - audit critical actions;
 - least privilege.
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24. PASSWORD / AUTH

Authentication provider manages secure password storage.

ARVENA never stores raw passwords.

Support: email/password initially.

Future: Google/Microsoft SSO.

25. SESSION SECURITY

On disable member: future requests denied immediately or within short token refresh window.

High-risk actions may require recent authentication later.

26. PRIVACY

Required product capabilities:

- data export;
 - account/organization deletion workflow;
 - retention policy;
 - consent-aware external messaging;
 - minimal worker/customer data exposure.
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27. OBSERVABILITY SLO

Alert on:

- elevated 5xx;
 - failed webhooks;
 - growing queue backlog;
 - failed storage uploads;
 - database connectivity;
 - auth degradation.
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28. ACCEPTANCE MATRIX — AUTH

Signup

Given valid email/password When signup completes Then onboarding starts.

Invalid Login

No sensitive account-existence leak beyond chosen auth UX.

Disabled Member

Cannot access organization records.

29. ACCEPTANCE MATRIX — ORGANIZATION

Owner creates organization.

All created tenant records contain same organization_id.

Member from other org cannot access them.

30. ACCEPTANCE MATRIX — CUSTOMER

Create customer: persists.

Normalize phone: works.

Duplicate phone: warning/prevention according to rule.

Archive: history remains.

Search: name and normalized phone work.

31. ACCEPTANCE MATRIX — LEAD

Create lead.

Assign sales.

Change status.

Schedule follow-up.

Convert to Job.

Lead history remains.

32. ACCEPTANCE MATRIX — QUOTE

Create quote.

Issue quote.

Snapshot items.

Accept.

Convert to Job.

Price catalog update does not alter issued quote.

33. ACCEPTANCE MATRIX — JOB

Create from scratch.

Create from lead.

Create from quote.

Assign visit.

Change priority.

Archive completed job only through allowed behavior.

34. ACCEPTANCE MATRIX — VISITS

One Job supports multiple Visits.

Completing one does not incorrectly complete Job.

Overlap detection works.

Timezone rendering correct.

Worker sees assigned visit only.

35. ACCEPTANCE MATRIX — WORKER

Worker: sees today.

Can Depart.

Can Arrive.

Can Start.

Can checklist.

Can upload photo.

Can Finish.

Cannot see unrelated Jobs.

36. ACCEPTANCE MATRIX — OFFLINE

Worker action offline: saved locally.

UI: Pending sync.

Reconnect: synchronizes once.

Duplicate action: not created.

37. ACCEPTANCE MATRIX — INVOICE

Draft invoice.

Issue.

Record partial payment.

Status PARTIALLY_PAID.

Record remaining payment.

Status PAID.

Historical price stable.

38. ACCEPTANCE MATRIX — FOLLOW-UP

Create.

Due today.

Visible on Today screen.

Snooze.

Complete.

Activity reflected.

39. ACCEPTANCE MATRIX — RECURRING

Monthly rule.

Generate near-term occurrence.

Do not generate endless years.

Change future rule.

Past completed Jobs unchanged.

Duplicate generator execution does not duplicate Job.

40. ACCEPTANCE MATRIX — WHATSAPP

Webhook with new phone: creates/link customer according to intake rule.

Duplicate webhook: one logical outcome.

Historical WhatsApp contacts: not mass-imported.

Provider outage: core system works.

41. ACCEPTANCE MATRIX — PERMISSIONS

Worker requests other org customer: deny.

Sales requests financial report without permission: deny.

Finance changes integration: deny.

Owner: allowed.

42. ACCEPTANCE MATRIX — IMPORT

Upload CSV.

Map columns.

Preview.

Detect invalid.

Detect duplicates.

Commit valid.

Import log created.

No silent partial corruption.

43. ACCEPTANCE MATRIX — EXPORT

Free organization can export basic business data.

Export contains only own organization data.

44. ACCEPTANCE MATRIX — RESPONSIVE

Each critical flow tested at:

375px mobile
768px tablet
1280px desktop

No inaccessible actions.

45. ACCEPTANCE MATRIX — ERROR HANDLING

Force network failure.

User input remains.

System explains failure.

Retry available.

No duplicate record after retry.

46. ACCEPTANCE MATRIX — CONCURRENCY

Open same invoice in two sessions.

Session A saves.

Session B saves stale copy.

Expected: conflict warning.

No silent overwrite.

47. P0 RELEASE BLOCKERS

P0 cannot release with:

- PlaceholderPage on core route;

- tenant data leak;
 - broken mobile worker flow;
 - duplicate submission bug;
 - invoice integrity bug;
 - missing auth guard;
 - missing RLS;
 - unusable onboarding;
 - no error/loading/empty state;
 - broken production build.
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48. PO DEFINITION OF DONE

A real service business can:

signup;

onboard;

create/import customer;

capture lead;

create quote;

convert to Job;

schedule multiple Visits;

assign worker;

worker executes from phone;

capture checklist/photos;

complete Job;

issue invoice;

record payment;

follow-up customer;

review dashboard/report;

export data.

Without spreadsheet being necessary for the core lifecycle.

49. P1 DEFINITION OF DONE

All P0 stable plus:

WhatsApp connector works reliably;

duplicate webhook safe;

automation template reliable;

custom fields work;

service catalog stable;

advanced recurring reliable;

team performance accurate;

usage controls enforced.

50. P2 DEFINITION OF DONE

All P1 stable plus:

multi-branch isolation;

custom roles;

approval;

full audit;

advanced automation;

SLA module;

public integrations;

higher-scale operational reporting.

51. IMPLEMENTATION ORDER

Do not build by sidebar order.

Build by dependency.

Foundation

Auth
Organization
Membership
Permissions
Design system
Tenant security

CRM

Customers
Leads
Follow-ups

Work

Jobs
Visits
Scheduling
Worker mobile

Evidence

Checklist
Photos
Notes
Timeline

Sales/Billing

Catalog
Quotes
Invoices
Payments

Retention

Assets
Recurring

Insights

Dashboard
Reports

Platform

Import/export
Settings
Billing/entitlements

Pro

WhatsApp
Automation

Ultra

Branches
Custom roles
Approval
Audit/SLA/API

52. BUILD RULE

No feature considered complete because: “the page exists”.

Complete means:

UI exists.

Backend rule exists.

Database constraint exists.

Permission exists.

Loading exists.

Empty state exists.

Error state exists.

Mobile works.

Acceptance test passes.

53. SOURCE OF TRUTH HIERARCHY

If implementation conflicts:

1. Master PRD ARVENA v1.0
2. This Specification Pack
3. Approved database migration
4. Approved UI implementation
5. Developer assumption

Developer assumption always loses to documented product rules.

54. CHANGE CONTROL

Material change requires decision record.

Format:

ADR-XXX

or

PRD v1.x

Examples requiring versioned change:

- merging Job and Visit;
 - removing export;
 - changing tier philosophy;
 - making WhatsApp mandatory;
 - redesigning tenancy;
 - changing invoice history behavior.
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55. FINAL BUILD-READY STATUS

After this specification:

Product scope: LOCKED

Core UX: LOCKED

Navigation: LOCKED

Critical database entities: LOCKED

Business states: LOCKED

Base permissions: LOCKED

Tier model: LOCKED

Background/event rules: LOCKED

Reliability baseline: LOCKED

Acceptance baseline: LOCKED

Implementation details that remain flexible:

- exact frontend library choices;
- exact component implementation;
- optimization strategy;
- vendor/provider choice where abstraction exists;
- minor visual polish.

These may evolve without changing ARVENA's product behavior.

FINAL PRINCIPLE

ARVENA must hide complexity from the user without hiding operational truth.

The backend can be sophisticated.

The owner's experience must remain simple:

What needs attention?

What is happening?

Who is responsible?

Has the customer paid?

What happens next?

That is the standard by which every ARVENA feature is accepted or rejected.